

NESNE TABANLI PROGRAMLAMA I

2025-2026 GÜZ

DÖNEM PROJESİ

Son Teslim Tarihi: **UZEM’de duyurulacaktır.**

Açıklama ve Kurallar:

1. Grup çalışması içerisinde,
 - a. Bütün grup üyeleri tasarım sürecinde bulunmalıdır.
 - b. Görevler (sınıf kodlarının ve uygulamanın yazılması) tüm grup üyelerine dağıtılmalı ve raporda bu dağılım açıkça belirtilmelidir.
 - c. Her öğrenci, kendisine atanan sınıf kodlarını ve test programlarını yazmalıdır. Her sınıf ayrı header (.h) ve ayrı source (.cpp) dosyasına sahip olmalıdır. Her dosyada kodu yazan kişinin ismi ve tarih bulunmalıdır.
2. Kodların yanında ilgili kod segmenti ile ilgili açıklamaların yazılması gerekmektedir.
3. Proje için tasarım ve gerçekleştirme aşamasının aktarıldığı bir rapor hazırlanması gerekmektedir.
4. Nesne tabanlı yapıların kullanıldığı (abstraction, inheritance, polymorphism, templates, exception handling, etc.) iyi bir tasarım ve kodlama yapmalısınız.
5. Proje için yazılan kodları ve proje raporunu sıkıştırarak .zip formatında “GrupNo.zip” şeklinde isimlendirip ESUZEM’e yüklemelisiniz. Gruptan bir kişinin yükleme yapması yeterlidir.
6. Uygulama programı konsol tabanlı olmalıdır.
7. Sadece standart C++ kütüphaneleri kullanılmalıdır.
8. Tasarım ve kodlarınızın başka gruplardan ve referans gösterilmemiş kaynaklardan alındığı belirlenirse puan kesintisi yapılacaktır.
9. Proje raporunuzu Türkçe veya İngilizce olarak hazırlayabilirsiniz.

PROJECT TITLE: Online Bookstore

Specifications:

In this project, you will be implementing simple software for an online bookstore in C++. The bookstore will provide a shopping experience for your customers, while at the same time allowing you to practice developing object-oriented coding as well as being a member of the team.

The class diagram which provides the framework for the bookstore is given in Figure 1. Here's a brief description of each class:

- **Product**
 - An abstract item for sale in the store
 - Notable attributes:
 - ID – unique ID for the item
 - name – the name of the item
 - price – the price of the item
 - Possible methods:
 - printProperties(): shows all properties of the items.
- **Book**
 - An item for sale in the store
 - Notable attributes:
 - author – the author of the book
 - publisher – the publisher of the book
 - page – the page number of the book
- **Magazine**
 - An item for sale in the store
 - Notable attributes:
 - issue – the issue of the magazine
 - type – the type of the magazine
- **MusicCD**
 - An item for sale in the store
 - Notable attributes:
 - singer – the name of the singer
 - type – the type of the music
- **Customer**
 - A customer in the store
 - Notable attributes:
 - customerID – unique ID for each customer
 - name – the name of the customer
 - bonus – each customer earns bonus as 1% of the amount of each shopping. Then, he/she may use his/her bonus next shopping or save it.
 - username and password – a customer may shop after entering his/her username and password.
- **ShoppingCart**
 - A shopping cart for the customer
 - Notable attributes:
 - productsToPurchase – the list of items to purchase currently in the shopping cart (use list or vector from STL)

- paymentMethod – the payment method to purchase items
 - isBonusUsed – this has the value false as a default. If this is set to true, the customer pays the purchase amount discounted by his/her bonus.
- Possible methods:
 - printProducts() – shows the items currently in the cart
 - addProduct(productToPurchase) – add an item to the shopping cart
 - removeProduct(productToPurchase) – remove an item from the shopping cart
 - placeOrder() – perform the payment for the items in the cart and send invoice to the customer's email.
 - cancelOrder() – cancel the order
 - showInvoice() – show the invoice
- ProductToPurchase
 - A class includes an item and its quantity for adding to cart.
 - Notable attributes:
 - product – an item to purchase
 - quantity – the quantity of the item to purchase
- Payment
 - An abstract for the payment methods
 - Notable attributes:
 - amount – an amount to pay
 - Possible methods:
 - performPayment() – authorize the payment
- Credit
 - A payment method
 - Notable attributes:
 - number – credit card number
 - type – type of the credit card (Master/Visa)
 - expDate – expiration date of the card
- Cash
 - A payment method
- Check
 - A payment method
 - Notable attributes:
 - name – name of the check owner
 - bankID – bank ID of the check

Note that required constructors and destructors are not listed in the above for the sake of brevity. You should add them.

Requirements:

Use the framework to write a working version of the online bookstore.

Your application program must have at least five customers and three products of each type.

Implement the user interface any way you like.

Feel free to add new classes or new data and methods for the given classes.

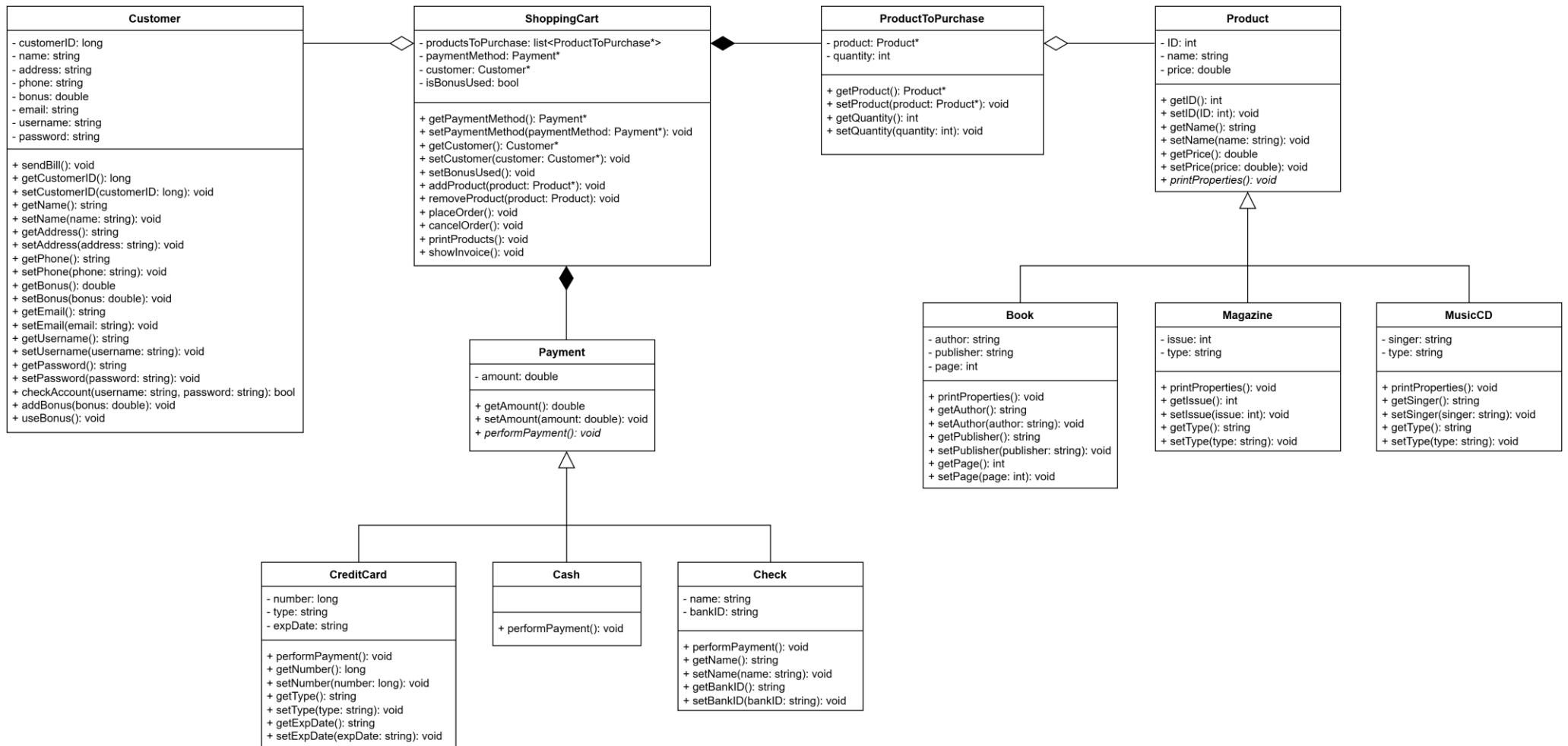


Figure 1. UML Class Diagram

Sample User Interface

You can also design the user interface menus as object for bonus points.

Main Menu

1. Customer
2. Item
3. Shopping
4. Quit

Choose one: 1 ↵

Customer Menu

1. Add a new customer to the system
2. Show the customers in the system
3. Back

Choose one: 1 ↵

Add Customer

Customer Name: Ali Kaya ↵

Customer Address: Batikent Mah. No: 4, Eskisehir ↵

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Customer Menu

1. Add a new customer to the system
2. Show the customers in the system
3. Back

Choose one: 3 ↵

Main Menu

1. Customer
2. Item
3. Shopping
4. Quit

Choose one: 3 ↵

Shopping Menu

1. Login
2. Add Product
3. Remove Product
4. List All Products
5. List Shopping Cart
6. Show Bonus
7. Use Bonus
8. Place Order
9. Cancel Order
10. Show Invoice
11. Quit

Choose one: 1 ↵

Enter the username: akaya ↵

Enter the password: 12345 ↵

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Report Contents

1. Introduction: General information about the project

2. Design: This part includes explanations about design, UML diagrams, task assignments, sample outputs of the application program for the sample inputs, etc.

Task Assignments (Must be included)

Group Member	Tasks
Name of group member	Class(es), menus, documentation, etc.
Name of group member	...
...	...

3. Conclusion: Evaluation of the project results, comments about teamwork, pros and cons, advice for further work, etc.



ESKİŞEHİR OSMANGAZİ ÜNİVERSİTESİ
MÜHENDİSLİK MİMARLIK FAKÜLTESİ

BİLGİSAYAR MÜHENDİSLİĞİ
NESNE TABANLI PROGRAMLAMA I

DÖNEM PROJESİ
ONLINE BOOK STORE

Öğretim Üyesi:
Dr. Öğr. Üyesi Serel AKYOL

Hazırlayan:
Ad Soyad – Öğrenci Numarası

Ay Yıl